HAOJIE HUANG

Email: huang.haoj@northeastern.edu
Personal Website: haojhuang.github.io

EDUCATION

Doctor of Philosophy | ***Computer Science*** — Sep. 2021 – Expected Dec. 2026
Northeastern University — Boston, MA

Master of Science | ***Robotics*** — Sep. 2018 – May 2021
Northeastern University — Boston, MA

Bachelor of Engineering | ***Mechanical Engineering*** — Sep. 2013 – May 2017
Shandong University — Jinan, China

RESEARCH INTERESTS

Robot Learning, Geometric Deep Learning, Manipulation, Symmetry / Equivariance, Action Representation, 2/3D Perception, Reinforcement Learning, Vision Language and Action Learning, Physics Simulators, Real-World Deployment, Agentic Robotics, Robot Harness

RESEARCH HIGHLIGHTS

20+ publications in leading robotics, machine learning, and computer vision journals and conferences, including IJRR, RSS, CoRL, RA-L, ICRA, ICML, ICLR, NeurIPS, and ECCV.

Symmetric Manipulation Learning: Pioneered the systematic study of task symmetries for fast, data-efficient manipulation policy learning, with original contributions to 3D grasping, pick-and-place, and multi-view closed-loop learning.

Geometric Action Representation: Introduced **Imagination Policy**, the first 3D generative world action model for manipulation, reformulating policy learning in a simpler, more natural geometric action space.

WORK EXPERIENCE

Applied Scientist Intern — Sep 2024 – Jan 2025
Amazon Robotics — North Reading, MA

- Worked on closed-loop Imitation learning algorithm

Research Scientist Intern — Sep 2023 – Jan 2024
Boston Dynamics AI Institute — Cambridge, MA

- Worked on the machine intelligence and robot learning algorithm

R&D Intern — June 2019 – Dec 2019
SharkNinja Corp. — Needham, MA

- Worked on the hardware(sensors, motors and etc.) and software(control algorithms, path planning and etc.) for robotic vacuums

TEACHING EXPERIENCE

Robotic Science and System (CS5335-Teaching Assistant) — Spring 2021
Northeastern University — Boston, MA

Technical Skills

Deep Learning & Distributed Computing: Neural network design and training for perception and policy learning with PyTorch; multi-GPU distributed training; Python multithreading and parallel data processing.

Simulation & Task Development: Custom task development across PyBullet, Drake, RLBench (PyRep), MuJoCo (robomimic, MimicGen), ManiSkill, and Isaac Sim.

Robot Hardware: Hands-on experience with UR, Franka, Kinova, xArm, bimanual systems (Piper-X), and robotic sensors.

Robotics Systems & Integration: ROS 2 and DORA; development of robotic systems integrating learned manipulation skills with planning.

Data & Training Infrastructure: Robotics data storage, cleaning, preprocessing, and transfer; streaming dataloaders; training monitoring.

Policy & System Evaluation: Design of evaluation protocols and performance metrics for manipulation policies and robotic systems.

Award

Best Paper Award on Compositional and Modular Learning in the Era of Scaling in Robotics Workshop of IROS 2026
Pittsburgh, United States — Sep 2026

Best Paper Award - Honorable Mention on IEEE Robotics and Automation Letters
Vienna, Austria — June 2026

Outstanding Paper Finalist at Conference on Robot Learning (CoRL) 2024
Munich, Germany — Nov 2024

Best Paper Award Finalist on Scaling Robot Learning Workshop of ICRA 2022
Philadelphia, United States — May 2022

Professional Contribution

2027 **Associate Editor:** ICRA; **Reviewer:** ICLR.

2026 **Reviewer:** ICLR, ICML, NeurIPS, ICRA, RSS, CoRL, RAL, TRO. **Invited Talks:** Tsinghua University.

2025 **Reviewer:** ICLR, ICML, NeurIPS, ICRA, IROS, RSS, CoRL, RAL, TRO. **Invited Talks:** Amazon Robotics. **Paper Presentations:** ICRA 2025.

2024 **Reviewer:** ICRA, IROS, RAL. **Invited Talks:** Worcester Polytechnic Institute (WPI); Next-Gen Robot Learning Symposium, TU Darmstadt. **Paper Presentations:** ICLR 2024, CoRL 2024.

2023 **Reviewer:** ICRA, IROS, RAL. **Paper Presentations:** ICRA 2023. **Program Committee:** Workshop on Symmetries in Robot Learning, RSS 2023.

2022 **Reviewer:** ICRA, IROS, RAL. **Paper Presentations:** ICRA 2022 (live demo and talk); RSS 2022; RLDM 2022.

First-Author Publications

Imitation Learning | Symmetric Manipulation Learning | Geometric Action Representation | Multi-View Geometry

Action Map Policy: Learning 3D Closed-loop Manipulation via Pixel Classification Jul 2026

Haojie Huang et al. ***(Contribution: first author)***

- *Arxiv Preprint -Under Submission*

Imitation Learning | Symmetric Manipulation Learning | Geometric Action Representation | Multi-View Geometry

Pix2Act: Image-Space Manipulation Policies with Equivariant Augmentation Jul 2026

Haojie Huang et al. ***(Contribution: first author)***

- *Arxiv Preprint -Under Submission*

Imitation Learning | Symmetric Manipulation Learning | Geometric Action Representation | 3D Perception and Vision

Match Policy: A Simple Pipeline from Point Cloud Registration to Manipulation Policies May 2025

Haojie Huang et al. ***(Contribution: first author)***

- *International Conference on Robotics and Automation, Atlanta, ICRA 2025*

Imitation Learning | VLA | Symmetric Manipulation Learning | Geometric Action Representation

Learning efficient and robust language-conditioned manipulation using textual-visual relevancy and equivariant language mapping June 2024

Mingxi Jia et al. ***(Contribution: co-first author)***

- *IEEE Robotics and Automation Letters 2025*
- *Best Paper Award Honorable Mention of IEEE Robotics and Automation Letters (5 out of 1700 publications)*

3D World Action Learning | Symmetric Manipulation Learning | Geometric Action Representation | 3D Perception and Vision

Imagination Policy: Using Generative Point Cloud Models for Manipulation Policies June 2024

Haojie Huang et al. ***(Contribution: first author)***

- *Conference on Robot Learning, Munich, CoRL 2024*

Imitation Learning | Symmetric Manipulation Learning | Geometric Action Representation | 3D Perception and Vision

Fourier Transporter: Bi-Equivariant Robotic Manipulation in 3D Sep 2023

Haojie Huang et al. ***(Contribution: first author)***

- *International Conference on Learning Representations, Vienna, ICLR 2024*

VLA | Symmetric Manipulation Learning | Geometric Action Representation

Language-conditioned Equivariant Grasp Sep 2023

Haojie Huang et al. ***(Contribution: first author)***

- *Arxiv Preprint*

Imitation Learning | Symmetric Manipulation Learning | Geometric Action Representation

Leveraging Symmetries in Pick and Place March 2023

Haojie Huang et al. ***(Contribution: first author)***

- *International Journal of Robotics Research, Volume 43, Issue 4, 2024*

Symmetric Manipulation Learning | Geometric Action Representation | 3D Perception and Vision | Sim2Real

Edge Grasp Network: A Graph-Based SE(3)-invariant Approach to Grasp Detection Dec 2022

Haojie Huang et al. ***(Contribution: first author)***

- *International Conference on Robotics and Automation, London, ICRA 2023*
- *Sim-to-Real Robot Learning, CoRL 22 workshop, Auckland*
- *Symmetries in Robot Learning, RSS 2023 workshop, Daegu*

Imitation Learning | Symmetric Manipulation Learning | Geometric Action Representation

Equivariant Transporter Network June 2022

Haojie Huang et al. ***(Contribution: first author)***

- *Robotics Science and System, New York, RSS 2022*
- *Scaling Robot Learning, ICRA 22 workshop, Philadelphia*, Best Paper Award Finalists
- *Reinforcement Learning and Decision Making, RLDM 2022, Rhode Island*

3D Perception and Vision

Graph Attention Shape Completion Network Dec 2021

Haojie Huang et al. ***(Contribution: first author)***

- *International Conference on 3D Vision, London, 3DV 2021*

Collaborated Publications

Agentic Robotics | Robot Harness

Retriever: Composing Closed-Loop Asynchronous Robot Programs Jul 2026

Linfeng Zhao et al. ***(Contribution: collaborator)***

- *Arxiv Preprint - under review*

VLA | Scaling Robot Learning

Scale Up Strategically: Learning Compositional Generalization via Bias-Aware Evaluation and Data Collection for Robotic Manipulation Jul 2026

Yu Qi et al. ***(Contribution: collaborator)***

- *Conference on Robot Learning, Austin, CoRL 2026*

3D Perception and Vision

Evaluating Reasoning Coherence in Video Generative Models with Text and Visual Hints March 2026

Yu Qi et al. ***(Contribution: collaborator)***

- *European Conference on Computer Vision - ECCV 2026*

Imitation Learning | Symmetric Manipulation Learning

Equivariant diffusion policy for sample-efficient robotic manipulation Feb 2026

Dian Wang et al. ***(Contribution: collaborator)***

- *International Journal of Robotics Research, IJRR 2026*

Imitation Learning | VLA

Generalizable Hierarchical Skill Learning via Object-Centric Representation Oct 2025

Haibo Zhao et al. ***(Contribution: collaborator)***

- *IEEE Robotics and Automation Letters 2026*

Reinforcement Learning

Accelerating Model-Based Reinforcement Learning Using Equivariance Sep 2025

Ahmed Agha et al. ***(Contribution: collaborator)***

- *Northeast Robotics Colloquium - NERC 2025*

3D Perception and Vision

Clebsch-Gordan Transformer: Fast and Global Equivariant Attention Sep 2025

Owen Howell et al. ***(Contribution: collaborator)***

- *Arxiv Preprint*

VLA Agentic Robotics

BEAR: Benchmarking and Enhancing Multimodal Language Models for Embodied AI Sep 2025

Yu Qi et al. ***(Contribution: senior collaborator)***

- *International Conference on Machine Learning - ICML 2026*

Reinforcement Learning Symmetric Manipulation Learning

Equivariant Goal Conditioned Contrastive Reinforcement Learning July 2025

Arsh Tangri et al. ***(Contribution: senior collaborator)***

- *Arxiv Preprint*

Imitation Learning Symmetric Manipulation Learning 3D Perception and Vision

3D Equivariant Visuomotor Policy Learning via Spherical Projection June 2025

Boce Hu et al. ***(Contribution: collaborator)***

- *Conference on Neural Information Processing Systems, San Diego , NeurIPS 2025*
- *NeurIPS Spotlight - One of 688 Spotlight papers among 21,575 main-track submissions*

Reinforcement Learning Symmetric Manipulation Learning Sim2Real

Push-Grasp Policy Learning Using Equivariant Models and Grasp Score Optimization April 2025

Boce Hu et al. ***(Contribution: collaborator)***

- *IEEE Robotics and Automation Letters 2025 - RAL 25*

Imitation Learning Symmetric Manipulation Learning Geometric Action Representation 3D Perception and Vision

Coarse-to-Fine 3D Keyframe Transporter Feb 2025

Xupeng Zhu et al. ***(Contribution: collaborator)***

- *Arxiv preprint*

Symmetric Manipulation Learning Geometric Action Representation 3D Perception and Vision Sim2Real

OrbitGrasp: SE(3)-Equivariant Grasp Learning June 2024

Boce Hu et al. ***(Contribution: co-second author)***

- *Conference on Robot Learning, Munich, CoRL 2024*

Imitation Learning Symmetric Manipulation Learning

Equivariant Diffusion Policy June 2024

Dian Wang et al. ***(Contribution: collaborator)***

- *Conference on Robot Learning, Munich, CoRL 2024, Outstanding Paper Finalist*

VLA Agentic Robotics

ThinkGrasp: A Vision-Language System for Strategic Part Grasping in Clutter June 2024

Yaoyao Qian et al. ***(Contribution: collaborator)***

- *Conference on Robot Learning, Munich, CoRL 2024*